

Scheme of Valuation/Answer Key			
(Scheme of evaluation (marks in brackets) and answers of problems/key)			
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY			
B.Tech Degree S7 (R, S) / S7 (PT) (R) Examination December 2023 (2019 Scheme)			
Course Code: EET401			
Course Name: ADVANCED CONTROL SYSTEMS			
Max. Marks: 100			Duration: 3 Hours
PART A			
		Answer all questions, each carries 3 marks.	Marks
1		Definition of phase variable – 1 mark Advantages – 2 marks	(3)
2		Choice of state variables – 1 mark $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} \dot{u}$ – 1 mark $y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ – 1 mark	(3)
3		Transfer function = $C(sI-A)^{-1}B + D$ – 1 mark Calculation – 1 mark Final answer – $2/(s+2)$ – 1 mark	(3)
4		$X(t) = e^{At}X_0$ – 1 mark Finding e^{At} – 1 mark $X(t) = e^{At} X_0 = \begin{bmatrix} e^t & 0 \\ te^t & e^t \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} e^t \\ te^t \end{bmatrix}$ – 1 mark	(3)

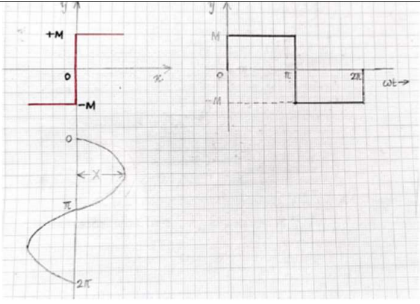
5		Definition of controllability – 1 mark Kalman's method with equations – 2 marks	(3)
6		Explanation of PBH test for observability – 1 marks. Equations – 2 marks	(3)
7		Any three non-linearities with figure – 1 mark each	(3)
8		Definition of limit cycle – 1 mark Stable limit cycle with figure – 1 mark Unstable limit cycle with figure – 1 mark	(3)
9		Definition of singular points – 1 mark Types of singular points with explanation – 2 marks.	(3)
10		(i) positive definite & example (1 mark) (ii) negative definite & example (1 mark) and (iii) indefinite & example (1 mark)	(3)
PART B			
<i>Answer any one full question from each module, each carries 14 marks.</i>			
		Module I	
11	a)	KVL equation - 1 mark Torque equation – 1 mark Eqn. for electro mechanical system – 1 mark Select suitable state variables and output variables, substitute them in the above three equations, rearrange the terms. (2 marks) Final state equation and Final output equation (1 mark) Block diagram – 1 mark	(7)
	b)	Rearranging the circuit – 1 mark	

		Node equations and solving for derivative of state variables – 4 marks	(7)
		$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -\frac{1}{RC} & \frac{1}{RC} \\ \frac{1}{RC} & -\frac{2}{RC} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{RC} \end{bmatrix} u$ - 1 mark $y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ - 1 mark	
		OR	
12	a)	Characteristic equation – 1 mark $\lambda^3 + 6\lambda^2 + 11\lambda + 6 = 0$ The eigenvalues are $\lambda = -1, -2, -3$ – 1 mark Modal matrix – 3 marks $M = [m_1 \ m_2 \ m_3] = \begin{bmatrix} 1 & 1 & 1 \\ -1 & -2 & -3 \\ 1 & 4 & 9 \end{bmatrix}$ Finding $A, \tilde{B}, \tilde{C}$ – 3 marks $\begin{bmatrix} \dot{z}_1 \\ \dot{z}_2 \\ \dot{z}_3 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} + \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} u \quad ; \quad Y = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix}$ - 2 marks	(10)
	b)	Any four advantages – 1 mark each	(4)
		Module II	
13	a)	Partial fraction expansion and finding residues – 1 marks Block diagram – 1 mark	(7)

		Finding state equations and output equation – 4 marks Final answer – 1 mark $\begin{bmatrix} x_1(k+1) \\ x_2(k+1) \\ x_3(k+1) \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} u$ $y = \begin{bmatrix} 2 & 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + 4u$	
	b)	Characteristic equation – 1 mark Eigen values – 1 mark $f(\lambda_i)$ – 1 mark $f(A)$ – 2 marks $A^k = \begin{bmatrix} 2(-1)^k - (-2)^k & (-1)^k - (-2)^k \\ -2(-1)^k + 2(-2)^k & -(-1)^k + 2(-2)^k \end{bmatrix}$ - 2 marks.	(7)
		OR	
14	a)	$X(t) = e^{At}X_0$ - 1 mark $e^{-At} X(t) = e^{-At} e^{At} X(0)$ $\therefore e^{-At} X(t) = X(0)$ - 1 mark Let $e^{-At} = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix}$ $\begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix} \begin{bmatrix} e^{-2t} \\ -2e^{-2t} \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$ - 1 mark	(10)

		$\begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix} \begin{bmatrix} e^{-t} \\ -e^{-t} \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ - 1 mark $\therefore e^{-At} = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix} = \begin{bmatrix} 2e^t - e^{2t} & e^t - e^{2t} \\ -2e^t + 2e^{2t} & -e^t + 2e^{2t} \end{bmatrix}$ $\therefore e^{At} = \begin{bmatrix} 2e^{-t} - e^{-2t} & e^{-t} - e^{-2t} \\ -2e^{-t} + 2e^{-2t} & -e^{-t} + 2e^{-2t} \end{bmatrix}$ - 2 marks $\mathbf{L}[e^{At}] = \phi(s)$ Where $\phi(s) = (sI - A)^{-1}$; $\therefore \phi(s)^{-1} = (sI - A)$ or $A = sI - \phi(s)^{-1}$ $\phi(s) = \mathbf{L}[e^{At}] = \begin{bmatrix} \frac{2}{s+1} - \frac{1}{s+2} & \frac{1}{s+1} - \frac{1}{s+2} \\ \frac{-2}{s+1} + \frac{2}{s+2} & \frac{-1}{s+1} + \frac{2}{s+2} \end{bmatrix}$ - 1 mark $\phi(s)^{-1} = \frac{1}{(s+1)(s+2)} \begin{bmatrix} s & -1 \\ 2 & s+3 \end{bmatrix} = \begin{bmatrix} s & -1 \\ 2 & s+3 \end{bmatrix}$ -2 marks $A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$ - 1 mark	
	b)	Proof of invariance of trace of a matrix	(4)
		Module III	
15	a)		(4)

		Explanation of state observers.	
	b)	State equation $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 10 \end{bmatrix} u$ - 3 marks $Q_c = [B \ AB \ A^2B] = \begin{bmatrix} 0 & 0 & 10 \\ 0 & 10 & -30 \\ 10 & -30 & 70 \end{bmatrix}$ - 2 marks Controllability checking - 1 mark Desired characteristic polynomial - 1 mark Determination of gain matrix $K = [0.4 \ 0.4 \ 1]$ - 3 marks	(10)
		OR	
16	a)	$Q_o = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$ - 2 marks; Determinant of $Q_o = 1$ - 1 mark Characteristic polynomial of original system, $\lambda^2 + 3\lambda + 1 = 0$ - 2 marks Desired characteristic polynomial, $\lambda^2 + 10\lambda + 25 = 0$ - 1 mark Observer gain matrix, $G = \begin{bmatrix} 7 \\ 10 \end{bmatrix}$ - 4 marks	(10)
	b)	Explanation of duality principle	(4)
		Module IV	

17	a)	 Input signal: $x = X \sin \omega t$ Output signal: $y(\omega t) = \begin{cases} +M & ; 0 < \omega t < \pi \\ -M & ; \pi < \omega t < 2\pi \end{cases}$ Describing function : $A_1 = \frac{4M}{\pi} \quad B_1 = 0 \quad \phi_1 = 0^\circ$ Derivation of constants with necessary steps $K_N(X, \omega) = \frac{Y_1}{X} \angle \phi_1 \quad Y_1 = \sqrt{A_1^2 + B_1^2} \quad \phi_1 = \tan^{-1} \left(\frac{B_1}{A_1} \right)$ $K_N(X, \omega) = \frac{4M}{\pi X} \angle 0^\circ$	(6)
	b)	Definition of Describing function – 2 marks Explanation of describing function analysis, concept of enclosure – 6 marks	(8)
		OR	
18	a)	Polar plot with $K = 1$ - 3 marks $-1/K_N$ locus - 2 marks Maximum value of K for the system to be stable ($K=2.25$) - 2 marks The frequency, $\omega = 1/\sqrt{2}$, amplitude = 1 and semi stable limit cycle - 3 marks	(10)
	b)	Any four properties of nonlinear systems.	(4)
		Module V	
19	a)	Determination of singular point – 3 marks Plot of phase trajectory (with necessary steps) - 6 marks Stable focus - 1 mark	(10)

	b)	Stable limit cycles - 2 marks Unstable limit cycles - 2 marks	(4)
		OR	
20	a)	Definition of each term - 1 mark each	(4)
	b)	$A^T P + P A = -I$ $\begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{bmatrix} + \begin{bmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{bmatrix} \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix} = - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ - 2 marks Solving for P $P = \begin{bmatrix} \frac{3}{2} & \frac{1}{2} \\ \frac{1}{2} & 1 \end{bmatrix}$ - 4 marks Positive definiteness of P for stability. $P_1 = \left \frac{3}{2} \right > 0 ; P_2 = \begin{vmatrix} \frac{3}{2} & \frac{1}{2} \\ \frac{1}{2} & 1 \end{vmatrix}$ $= \frac{3}{2} - \frac{1}{4} = \frac{6}{4} - \frac{1}{4} = \frac{5}{4} > 0$ - 2 marks Lyapunov function V(x) $V(x) = x^T P x$ $V(x) = \left(\frac{3}{2}\right)x_1^2 + x_1 x_2 + x_2^2$ - 2 marks	(10)

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